

Grade 2 Ontario Curriculum Data

Data Detectives Return

Unit Overview: Data Detectives Return

Grade: Grade 2 | **Strand:** Data

Target Expectations: D1.1, D1.2, D1.3, D1.4, D1.5

Learning Goal: Children will sort by 2 attributes using Venn/Carroll, collect 2-way tally tables, build bar graphs, find the mode, and write conclusions.

Lesson	Lesson Title	Learning Goal
Day 1	Sort by Two Rules	I can sort by two attributes using Venn or Carroll.
Day 2	Two-Way Tally Tables	I can make a two-way tally table.
Day 3	Bar Graph Builder	I can build a bar graph with title and labels.
Day 4	What's the Mode?	I can find the mode of a data set.
Day 5	The Bar Graph Case Capstone	I can analyse a bar graph and write a conclusion using mode.

Lesson 1: Sort by Two Rules

Goal: We can sort by TWO rules at once. Venn and Carroll diagrams show overlap.

Expectation: D1.1

Learning Goal: "I can sort by two attributes using Venn or Carroll."

Materials: M1_venn_mat, M2_carroll_mat, M5_data_cards, M7_data_chart

1. Minds On (12 mins): Detective Dot Sorts Two Ways

Detective Dot: This time I sort by TWO rules! Red AND square — overlap! Carroll table — red AND not-square... Detective Dot opens: 'BOTH? Only one? Or NEITHER?' (Class echoes.)

Bring children to the carpet. Detective Dot and Tally puppets. 'Friends — last year you sorted by ONE rule. Today we sort by TWO RULES at once.'

2. Action (25 mins): Sort Stations

- Whole group: sort 4 cards by 2 attributes (red/blue × square/triangle).
- Three stations.
- Stretch challenge at Station C: 3-attribute sort.
- Discuss when to use Venn vs Carroll.
- Circulate to record diagnostic evidence.
- STRATEGY COMPARE: Two pairs share their Venn vs Carroll sorts of the same set.

3. Consolidation (13 mins): Detective Dot's Two-Sort Brag

Discuss:

- What's a Venn diagram?
- What's a Carroll diagram?
- When is Venn better?
- When is Carroll better?
- Dot's preview: tomorrow Tally turns today's two-attribute thinking into a TWO-WAY TALLY TABLE (Lesson 2 — D1.1).

Exit: Detective Dot's 4-zone refrain — class chants: 'BOTH? Only one? Or NEITHER?' (3x).

Lesson 2: Two-Way Tally Tables

Goal: Two-way tally tables ROW-by-ROW capture TWO PIECES of information per data point.

Expectation: D1.2

Learning Goal: "I can make a two-way tally table."

Materials: M3_two_way_table, M9_question_cards, M7_data_chart, M10_unit_poster

1. Minds On (12 mins): Tally Builds a Two-Way Table

Tally: I asked TWO questions: favourite snack AND favourite drink. Now I record BOTH answers per child! Dot opens: 'BOTH? Only one? Or NEITHER?'
Tally adds: 'Row, Column, Cell!'

Bring children to the carpet. Detective Dot and Tally puppets. 'Friends — yesterday Venn and Carroll. Today TWO-WAY TALLY TABLES.'

2. Action (25 mins): Two-Way Tally Stations

- Whole group: tally 6 children's snack + drink choices in 2x2 table.
- Three stations.
- Stretch at Station C: 3x3 table.
- Discuss row vs column totals.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two pairs share their two-way tables for the same survey data.

3. Consolidation (13 mins): Tally's 2-Way Brag

Discuss:

- What's a two-way tally?
- What does each CELL mean?
- What's the row total?
- What's the column total?

Exit: Detective Dot's table refrain — class chants: 'Row says ____.'

Lesson 3: Bar Graph Builder

Goal: A BAR GRAPH uses bars instead of pictures. The bar HEIGHT shows the count.

Expectation: D1.3

Learning Goal: "I can build a bar graph with title and labels."

Materials: M4_bar_graph, M7_data_chart, M10_unit_poster

1. Minds On (12 mins): Detective Dot Builds a Bar

Detective Dot: A pictograph uses pictures. A BAR GRAPH uses bars! Each bar's HEIGHT is the count! Dot opens: 'Bar height equals...' Class: 'the COUNT!'

Bring children to the carpet. Detective Dot and Tally puppets. 'Friends — yesterday two-way tallies. Today BAR GRAPHS.'

2. Action (25 mins): Bar Graph Stations

- Whole group: build a 4-category bar graph from class snack data.
- Three stations.
- Stretch at Station C: 5+ categories.
- Discuss the title-and-labels rule.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two pairs share their bar graphs for the same two-way table.

3. Consolidation (13 mins): Detective Dot's Bar Brag

Discuss:

- What's a BAR GRAPH?
- What does bar HEIGHT mean?
- Where does the TITLE go?
- What do AXIS LABELS tell us?

Exit: Detective Dot's bar-graph refrain — Dot calls 'Bar height equals...' Class: 'the COUNT!' Repeat 3x.

Lesson 4: What's the Mode?

Goal: The MODE is the value that appears MOST often. Some data sets have NO mode or MORE THAN ONE.

Expectation: D1.4

Learning Goal: "I can find the mode of a data set."

Materials: M4_bar_graph, M7_data_chart, M8_mode_chart, M10_unit_poster

1. Minds On (12 mins): Tally Finds the Mode

Tally: This bar graph shows pets. DOG bar is HIGHEST. Dog is the MODE — most common! Dot opens: 'Most counts wins — that's the MODE!'

Bring children to the carpet. Detective Dot and Tally puppets. 'Friends — yesterday bar graphs. Today MODE.'

2. Action (25 mins): Mode Stations

- Whole group: find the mode in 4 different displays.
- Three stations.
- Stretch at Station C: find a 'NO MODE' or 'TWO MODES' example.
- Discuss bimodal and no-mode cases.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two pairs share how they found the mode on the same graph — Pair A 'I read all heights, picked tallest'; Pair B 'I scanned for the longest bar, then read the...

3. Consolidation (13 mins): Tally's Mode Brag

Discuss:

- What is the MODE?
- When is there NO mode?
- Can there be TWO modes?
- Why do detectives care about mode?

Exit: Detective Dot's mode refrain — class chants: 'Most counts wins — that's the MODE!' Each child reads one graph's mode aloud with the sentence frame.

Lesson 5: The Bar Graph Case Capstone

Goal: Detectives ANALYSE bar graphs and write CONCLUSIONS. The MODE tells what's most popular.

Expectation: D1.5

Learning Goal: "I can analyse a bar graph and write a conclusion using mode."

Materials: M1_venn_mat, M2_carroll_mat, M3_two_way_table, M4_bar_graph

1. Minds On (12 mins): Detective Dot Solves the Case

Detective Dot: 5 detective steps! ASK + SURVEY + GRAPH + MODE + CONCLUDE!

Bring children to the carpet. Detective Dot and Tally puppets in detective hats. 'Friends — today is our CAPSTONE. We use ALL 5 detective skills.'

2. Action (25 mins): The Detective Case

- Hand out templates.
- Step 1-2 (5 min): Pick question + tally 8 classmates.
- Step 3 (8 min): Build a bar graph.
- Step 4 (5 min): Find the mode.
- Step 5 (5 min): Conclude.

3. Consolidation (13 mins): Detective Dot's Case Brag

Discuss:

- What's the mode of your data?
- What's a CONCLUSION?
- How does the data tell the story?
- What was your favourite step?

Exit: Each child shares ONE conclusion: 'My mode was ____.'

Name: _____

Date: _____

Detective Dot's Two-Way Sort

Learning Goal: I can sort by two attributes using Venn or Carroll.

Instructions for the Student:

1. Place 6 items in the Venn (red circle / square circle / both / neither).
2. Place the same 6 items in a Carroll (red x square).
3. Pick 2 attributes for 4 of your toys.

My Sort

Make your own Venn or Carroll:

Name: _____

Date: _____

Tally's Two-Way Survey

Learning Goal: I can make a two-way tally table.

Instructions for the Student:

1. Read the table.
2. Survey 6 classmates with 2 questions.
3. Make your own 2-way survey at home (3 family members).

Make Your Own Table

	Col 1	Col 2	Total
Row A			
Row B			
Total			

Name: _____

Date: _____

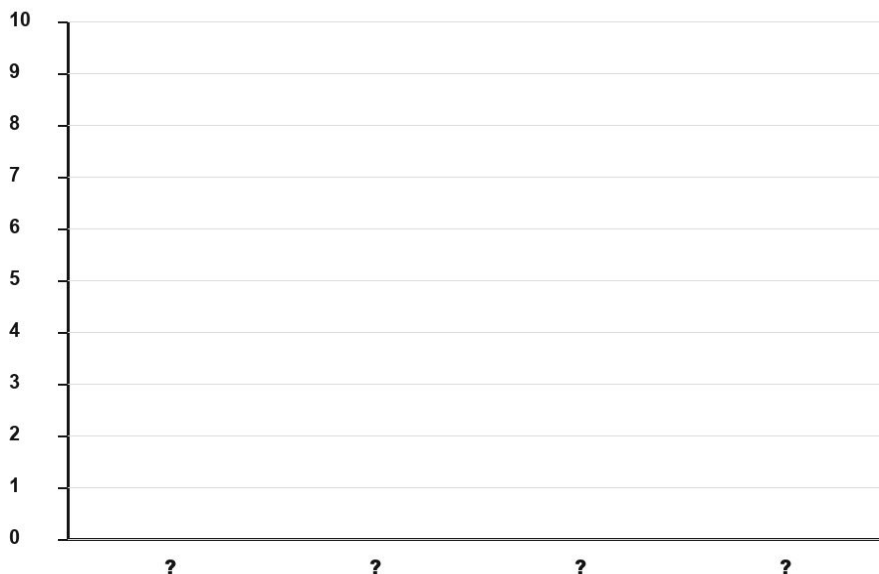
Detective Dot's Bar Builder

Learning Goal: I can build a bar graph with title and labels.

Instructions for the Student:

1. Read the bar graph.
2. Use the tally to build a bar graph.
3. Survey 5 classmates.

My Bar Graph



Name: _____

Date: _____

Tally's Mode Hunt

Learning Goal: I can find the mode of a data set.

Mission

1. Find the mode in 4 displays.
2. Look at 2 special data sets.
3. Make a data set with a clear mode.

Special Categories

The most common.

MODE

Highest minus lowest.

RANGE

A value far from the rest.

OUTLIER

Name: _____

Date: _____

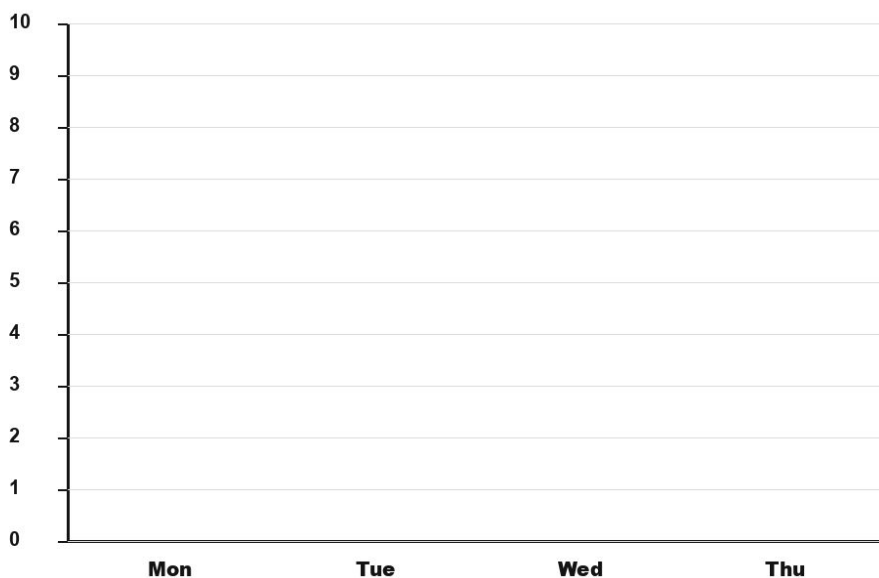
Detective Dot's Final Case

Learning Goal: I can analyse a bar graph and write a conclusion using mode.

Instructions for the Student:

1. Pick 2 questions.
2. Build a bar graph from your tally.
3. Write a conclusion using the mode.

Capstone Bar Graph

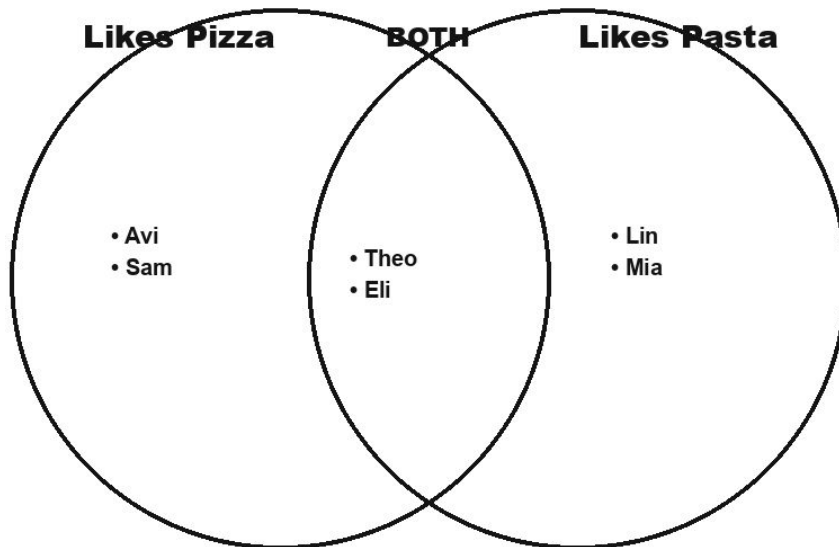


Venn Diagram Mat

(M1_venn_mat)

Letter-size mat with 2 overlapping circles for 2-attribute sorting.

Likes Pizza / Likes Pasta



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 12 mats.
2. Laminate for dry-erase.
3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 mats
Lessons: 1, 5
Prep: ~15 min

Carroll Diagram Mat

(M2_carroll_mat)

Letter-size 2x2 mat with yes/no headers.

Carroll Diagram

	Walks to school	Doesn't walk
Pet	• Sam • Lin	• Avi
No Pet	• Theo	• Mia • Eli

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 mats.

2. Laminate.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 mats

Lessons: 1, 5

Prep: ~15 min

Two-Way Tally Template

(M3_two_way_table)

Letter-size 2-way tally chart template.

Two-Way Frequency Table

	Walk	Bus	Car	Total
Boys	5	3	2	
Girls	4	4	2	
Total				

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 templates.

2. Laminate.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 templates

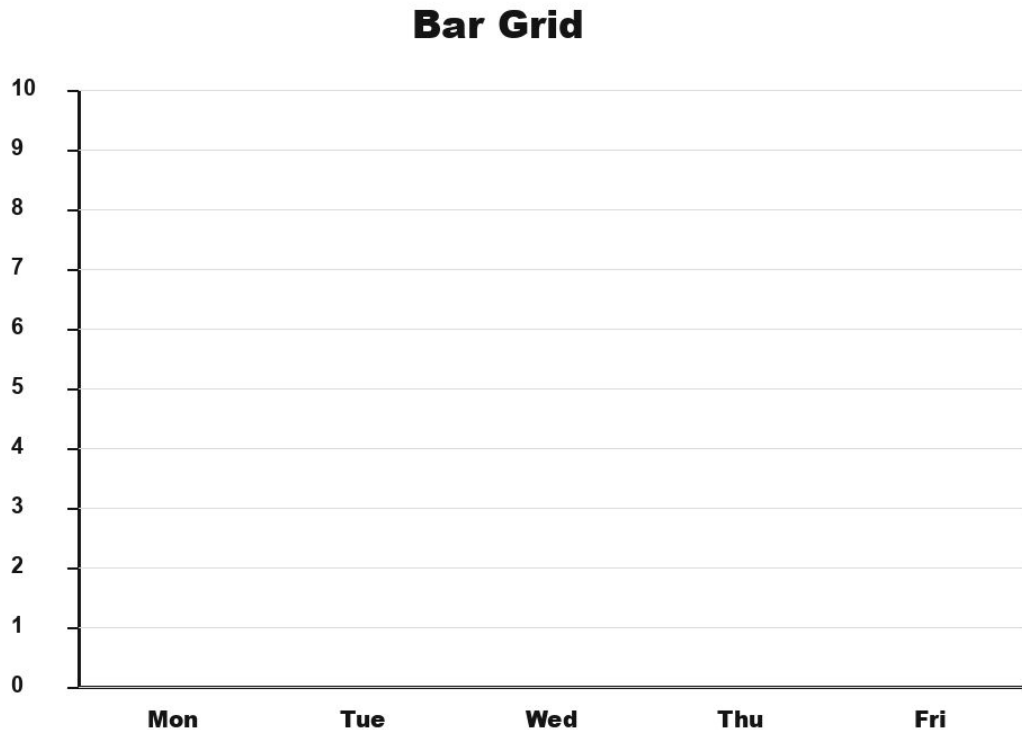
Lessons: 2, 5

Prep: ~15 min

Bar Graph Grid

(M4_bar_graph)

Letter-size bar graph grid with axes.



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 12 grids.
2. Laminate.
3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 grids
Lessons: 3, 4, 5
Prep: ~15 min

Survey Data Cards

(M5_data_cards)

Set of 30 paper data cards with 2 attributes each (e.g., colour + shape).



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 30 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 2, 5

Prep: ~25 min

Detective Case Template

(M6_capstone_template)

Letter-size capstone template with all 5 detective steps.

1. My QUESTION: _____ _____ _____
2. My DATA (tally): _____ _____ _____
3. My PICTOGRAPH: _____ _____ _____
4. What I FOUND: _____ _____ _____
5. My CONCLUSION: _____ _____ _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates.

2. Stack near teacher table.

3. No lamination.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

Data Vocabulary Chart

(M7_data_chart)

Anchor chart of new G2 data vocabulary.

Data Words	
DATA	Information we collect to answer a question.
TALLY	Marks we use to count. Five marks = a bundle.
PICTOGRAPH	A graph that uses pictures to show data.
CATEGORY	A group that things belong to.
MOST	The category with the largest number.
LEAST	The category with the smallest number.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

What is Mode? Chart

(M8_mode_chart)

Anchor chart explaining mode with examples.

MODE	
MODE	The most common value in a data set.
Find it	Look for the value that appears most often.
Example	1, 2, 2, 3, 4 → mode = 2
Tip	There can be more than one mode.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape next to vocab chart.

USE IN CLASS

Quantity: 1 chart

Lessons: 4, 5

Prep: ~10 min

Two-Question Survey Cards

(M9_question_cards)

Set of 12 survey cards each with 2 questions.

Pizza or pasta? ☐ ☐ <input type="triangle"/>	Walk or bus? ☐ ☐ <input type="triangle"/>	Cat or dog? ☐ ☐ <input type="triangle"/>
Sun or rain? ☐ ☐ <input type="triangle"/>	Apple or banana? ☐ ☐ <input type="triangle"/>	Math or art? ☐ ☐ <input type="triangle"/>
Read or play? ☐ ☐ <input type="triangle"/>	Snow or sand? ☐ ☐ <input type="triangle"/>	Day or night? ☐ ☐ <input type="triangle"/>

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 5

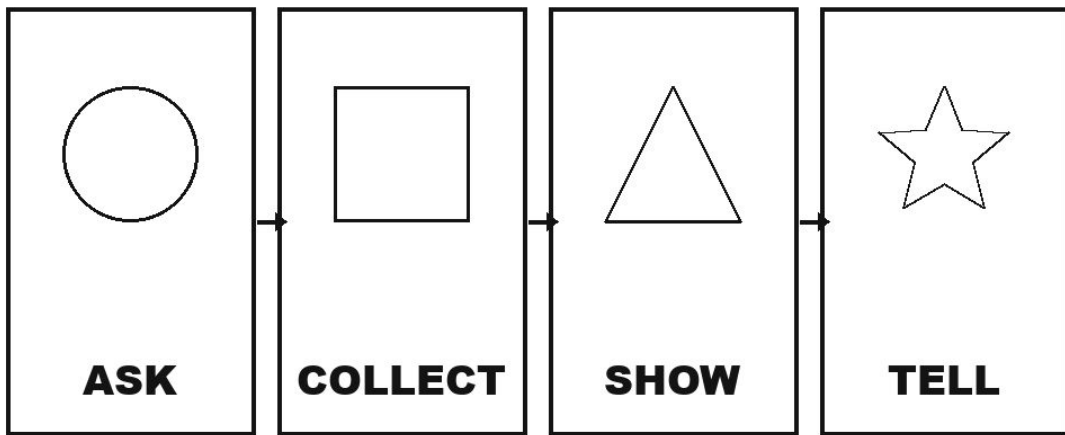
Prep: ~15 min

Data Detectives Return Poster

(M10_unit_poster)

Unit anchor poster of Dot and Tally with G2 detective steps.

Data Detectives!



Ask. Collect. Show. Tell.

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to front wall on Day 1.

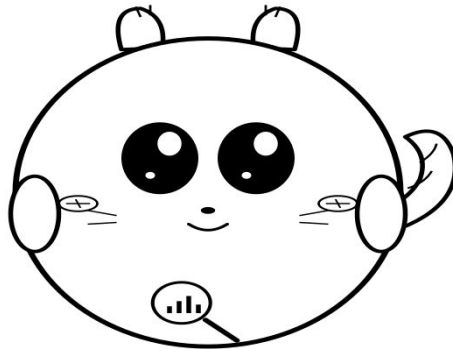
USE IN CLASS

Quantity: 1 poster
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

Detective Dot Puppet

(char_dot_puppet)

Paper puppet of Detective Dot for teacher use in all 5 lessons.



Detective Dot the Squirrel

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

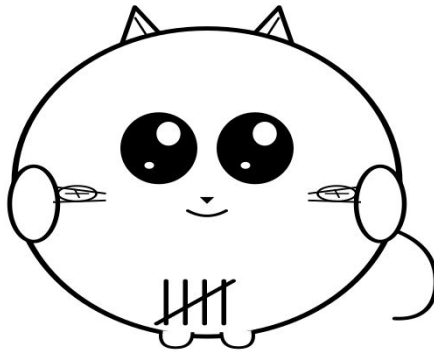
Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

Tally Puppet

(char_tally_puppet)

Paper puppet of Tally for teacher use in Lessons 2-5.



Tally the Tally Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~12 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
D1.1 sort sets of data about people or things according to two attributes, using tables and logic diagrams, including Venn and Carroll	Sorts by one attribute only. Cannot use Venn or Carroll.	Sorts by 2 attributes with prompts using either Venn OR Carroll.	Independently sorts by 2 attributes using both Venn and Carroll diagrams correctly.	Sorts by 3 attributes (3-circle Venn) and explains when to use Venn vs Carroll.
D1.2 collect data through observations, experiments, and interviews to answer questions of interest that focus on two pieces of	Cannot ask 2 questions independently. Tally in wrong cell.	Asks 2 questions with prompts. Tally with single-cell rule with prompts.	Independently asks 2 questions, surveys 6+, tallies 1-cell-per-child correctly.	Surveys 10+, totals rows + columns accurately.
D1.3 display sets of data, using one-to-one correspondence, in concrete graphs, pictographs, line plots, and bar graphs with proper	Bar heights don't match counts. No title or labels.	Bar heights mostly match. Title or labels missing.	Bar heights match exactly. Title and labels both present.	Bar graph + line plot or pictograph (multiple displays).
D1.4 identify the mode(s), if any, for various data sets presented in concrete graphs, pictographs, line plots, bar graphs, and	Cannot identify the mode.	Identifies single mode with prompts.	Independently identifies mode in bar graphs and tally tables. Recognizes no-mode and bimodal cases.	Explains why mode matters: 'mode tells us what's most popular'.
D1.5 analyse different	Personal preference instead of data	Group conclusion with sentence frame.	Independent group conclusion using	Comparison conclusion +

Junior Data Detective Certificate (G2)

This certificate is proudly presented to:

For successfully completing the 5-Day Data Detectives Return unit and demonstrating the skills of a G2 Junior Data Detective. A G2 data certificate. Detective Dot the Sorting Squirrel congratulates the student.

By completing this unit, this student has demonstrated:

- Sorting by 2 attributes with Venn and Carroll
- Building 2-way tally tables
- Constructing bar graphs
- Finding the mode and writing conclusions

Marketplace Listing

A 5-day Grade 2 unit on data framed around Detective Dot the Sorting Squirrel and Tally the Tally Cat.

Price: \$8.99 CAD

Pages: ~45

Classroom time: 250 min

Teacher prep: ~200 min

Top tags: Grade 2, math, data, venn diagram, carroll diagram, two-way tally, bar graph, mode

What's included:

- Unit blueprint with curriculum mapping (1 file)
- 5 lesson plans (5 files)
- 5 student worksheets (5 files)
- Manipulatives plan with 12 print-ready assets (1 file)
- Formative + reflection sheets (1 file)
- Assessment suite (1 file)